

Introduction to the tidyverse

Scientific workflows: Tools and Tips 

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What is this lecture series?

Scientific workflows: Tools and Tips

 Every 3rd Thursday  4-5 p.m.  Webex

- One topic from the world of scientific workflows
- Material provided [online](#)
- If you don't want to miss a lecture
 - [Subscribe to the mailing list](#)

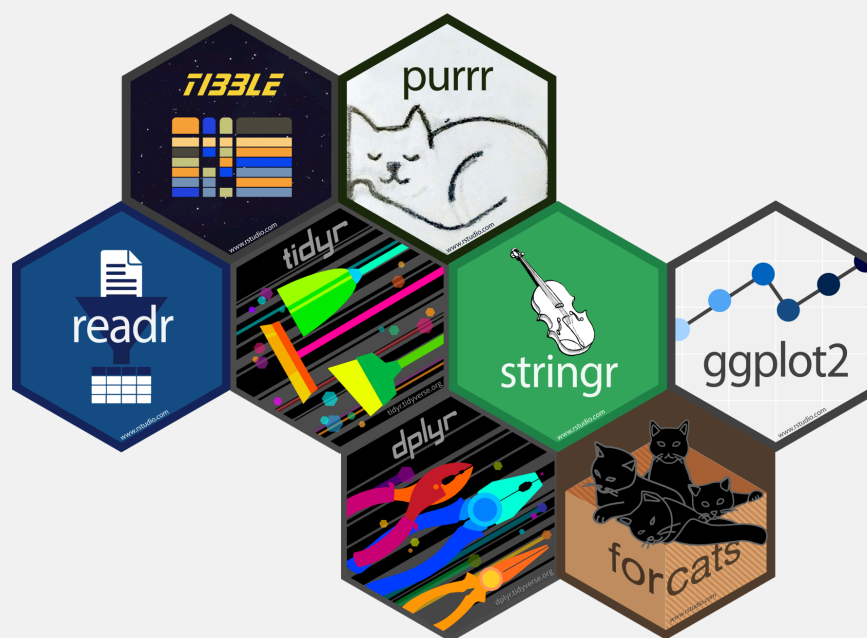
What is the tidyverse?



The tidyverse is an opinionated collection of R packages designed for data science. All packages share an underlying design philosophy, grammar, and data structures.

www.tidyverse.org

Tidyverse core packages



Why the tidyverse?



Basic idea: Make data analysis **efficient** and **intuitive**

- Write code that is easy to **read**, **write** and **maintain**
- More time for **data interpretation**
- Tidyverse is **actively developed** and has a **large community**.

Today



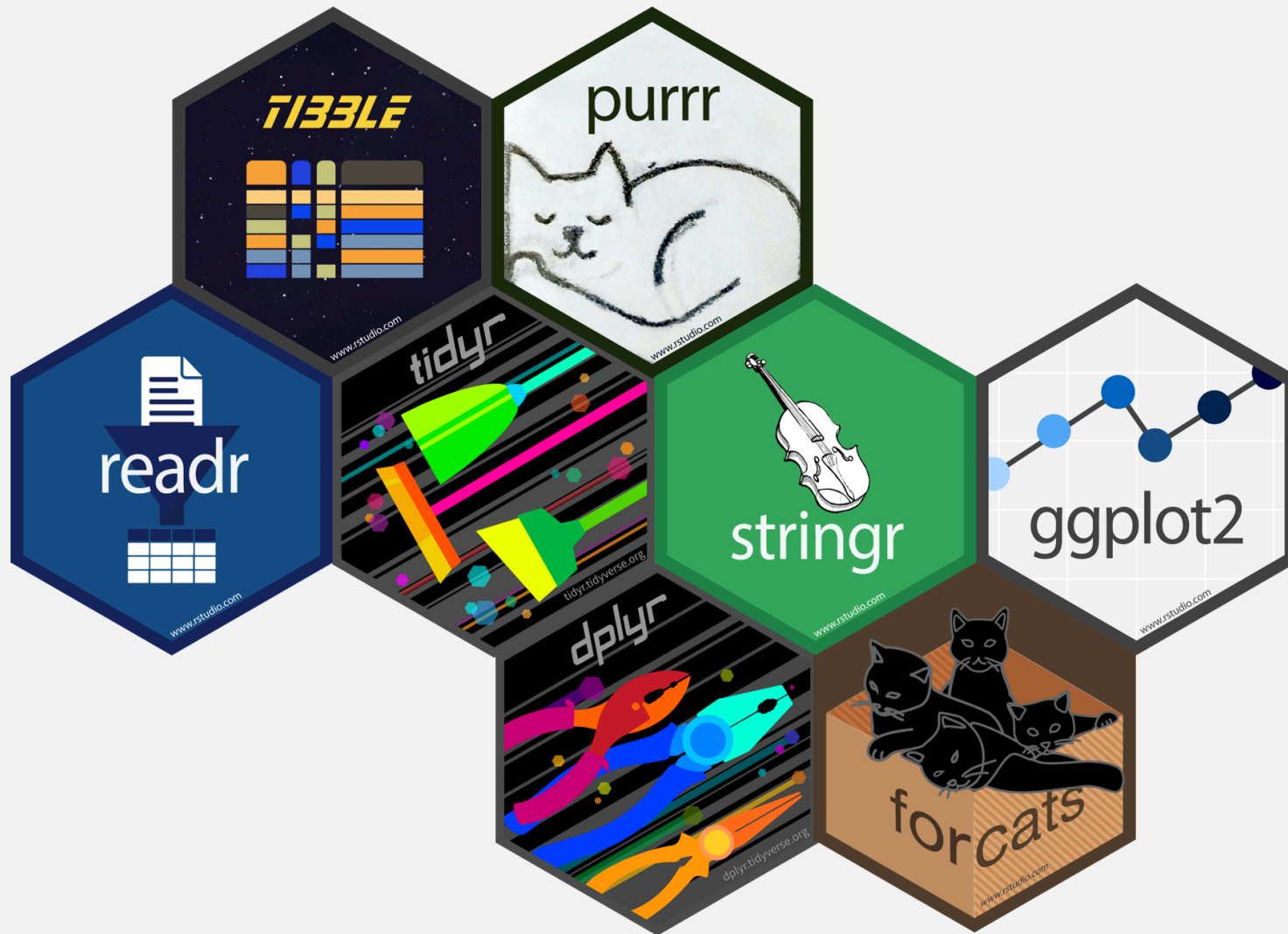
Overview of

- most important **packages** and **functions**
- how they work together in a **data analysis workflow**
- underlying **principles** of the tidyverse

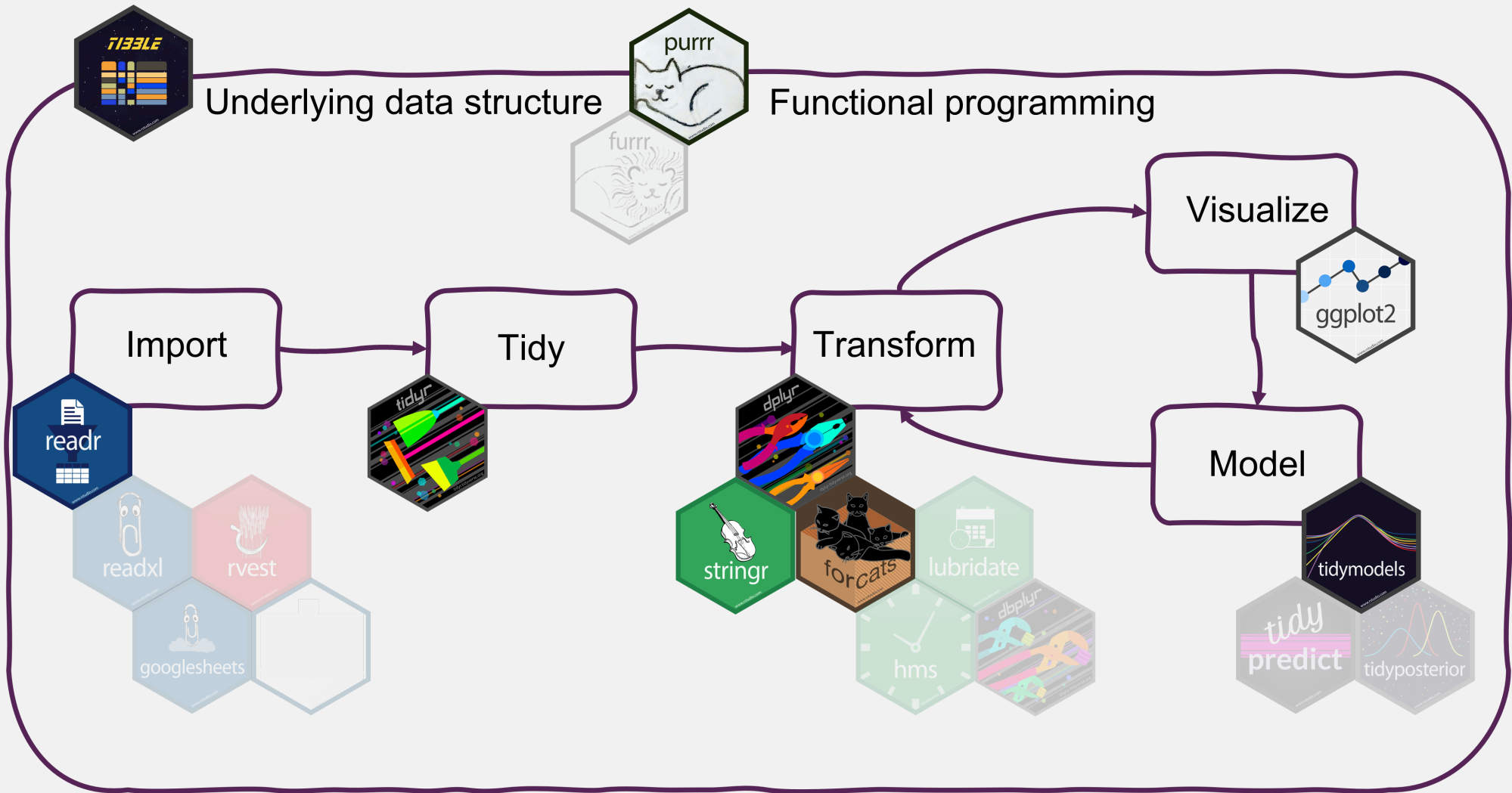
I can't show you everything. But the tidyverse has an **excellent documentation** and **cheatsheets**.

Which tidyverse packages do you use?

Head over to the menti poll



Data analysis with the tidyverse



Getting the tidyverse

Install all tidyverse packages with

```
install.packages("tidyverse")
```

Load all core tidyverse packages with

```
library(tidyverse)
#> — Attaching core tidyverse packages — tidyverse 2.0.0 —
#> ✓ dplyr      1.1.4      ✓ readr      2.1.5
#> ✓ forcats    1.0.0      ✓ stringr    1.5.1
#> ✓ ggplot2     3.5.1      ✓ tibble     3.2.1
#> ✓ lubridate  1.9.4      ✓ tidyr      1.3.1
#> ✓ purrr      1.0.2
#> — Conflicts — tidyverse_conflicts() —
#> ✗ dplyr::filter() masks stats::filter()
#> ✗ dplyr::lag()    masks stats::lag()
#> ⓘ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

You can also load packages individually - But don't do both!

Non-core packages need to be loaded individually, e.g.

```
library(readxl)
```

Tibbles from `tibble`

The underlying data structure



Underlying data structure

What are tibbles?



Tibbles are

a **modern reimagining** of the data frame. Tibbles are designed to be (as much as possible) **drop-in replacements** for data frames.

(Wickham, [Advanced R](#))

- Tibbles have the **same basic properties** as data frames
- Everything that you can do with data frames, you can do with tibbles
- Main advantage: Tibbles **print much nicer** to the console

Have a look at [this book chapter](#) for a full list of the differences between data frames and tibbles and the advantages of using tibbles.

How to create tibbles?



- Tidyverse functions return tibbles by default
- Create with `tibble` function (equivalent to `data.frame`)

How to create tibbles?



Creating a tibble

```
tibble(  
  x = 1:26,  
  y = letters,  
  z = factor(LETTERS)  
)  
#> # A tibble: 26 × 3  
#>       x y      z  
#>   <int> <chr> <fct>  
#> 1     1 a      A  
#> 2     2 b      B  
#> 3     3 c      C  
#> 4     4 d      D  
#> 5     5 e      E  
#> 6     6 f      F  
#> 7     7 g      G  
#> 8     8 h      H  
#> 9     9 i      I  
#> 10    10 j      J  
#> # i 16 more rows
```

Creating a data frame

```
data.frame(  
  x = 1:26,  
  y = letters,  
  z = factor(LETTERS)  
)  
#>       x y z  
#> 1     1 a A  
#> 2     2 b B  
#> 3     3 c C  
#> 4     4 d D  
#> 5     5 e E  
#> 6     6 f F  
#> 7     7 g G  
#> 8     8 h H  
#> 9     9 i I  
#> 10    10 j J  
#> 11    11 k K  
#> 12    12 l L  
#> 13    13 m M  
#> 14    14 n N  
#> 15    15 o O
```

Import data with readr

Read in your files





Most important **readr** functions

- **read_csv** to read **comma delimited** files
- **read_tsv** to read **tab delimited** files
- **read_delim** to read files with **any delimiter**

All **read_*** functions return a **tibble**.

Example with arctic temperature data



Data modified from [lterdatasampler](#)

```
arc_weather <- read_csv(file = "data/arc_weather.csv")
arc_weather
#> # A tibble: 365 × 3
#>   date          foolik_mean_temp_c poolik_mean_temp_c
#>   <date>          <dbl>          <dbl>
#> 1 1988-06-01          8.4          11.8
#> 2 1988-06-02           6           8.90
#> 3 1988-06-03          5.8          8.14
#> 4 1988-06-04          1.8          3.95
#> 5 1988-06-05          6.8          9.81
#> 6 1988-06-06          5.2          8.74
#> 7 1988-06-07          2.2          6.11
#> 8 1988-06-08          9.4         12.1
#> 9 1988-06-09         13.1         16.5
#> 10 1988-06-10         17.7         20.9
#> # i 355 more rows
```

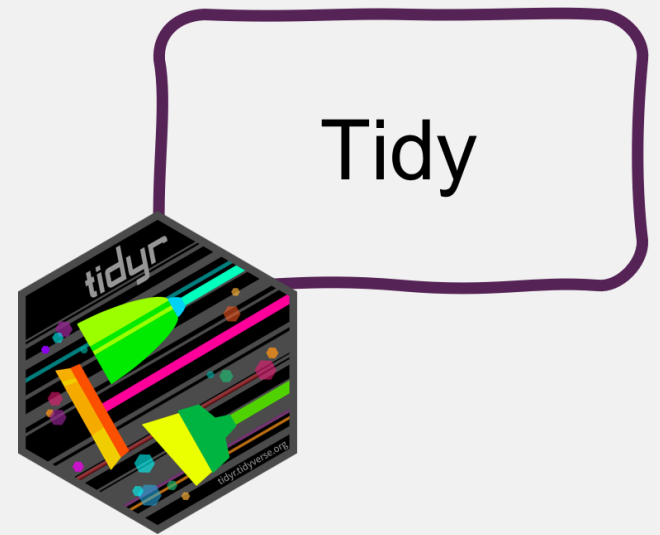


Advantages of readr

- Faster than `read.table/read.csv`
- Better defaults than base R (e.g. guessing of data types)
- Returns tibbles
- Tidyverse packages for other data types:
 - `readxl` for Excel files
 - `haven` for SPSS, SAS, and Stata files
 - `googlesheets4` for Google sheets
 - ...

Tidy data with `tidyr`

Reorganize the data



What is tidy data?

“**TIDY DATA** is a standard way of mapping the meaning of a dataset to its structure.”

—HADLEY WICKHAM

In tidy data:

- each variable forms a column
- each observation forms a row
- each cell is a single measurement

each column a variable

id	name	color
1	floof	gray
2	max	black
3	cat	orange
4	donut	gray
5	merlin	black
6	panda	calico

each row an observation

Wickham, H. (2014). Tidy Data. Journal of Statistical Software 59 (10). DOI: 10.18637/jss.v059.i10

Illustration from the [Openscapes](#) blog *Tidy Data for reproducibility, efficiency, and collaboration* by Julia Lowndes and Allison Horst

What is non-tidy data?

Tidy ::: {.cell} ::: {.cell-output-display}

id	name	color
1	floof	gray
2	max	black
3	cat	orange
4	donut	gray
5	merlin	black
6	panda	calico

...

Non-tidy ::: {.cell} ::: {.cell-output-display}

floof	max	cat	donut	merlin	panda
gray	black	orange	gray	black	calico

::: {.cell} ::: {.cell-output-display}

gray	black	orange	calico
floof	max	cat	panda
donut	merlin		

:::
 :::
 :::
 :::

Advantages of tidy data

The main advantages of **tidy** data is that the **tidyverse** packages are built to work with it.

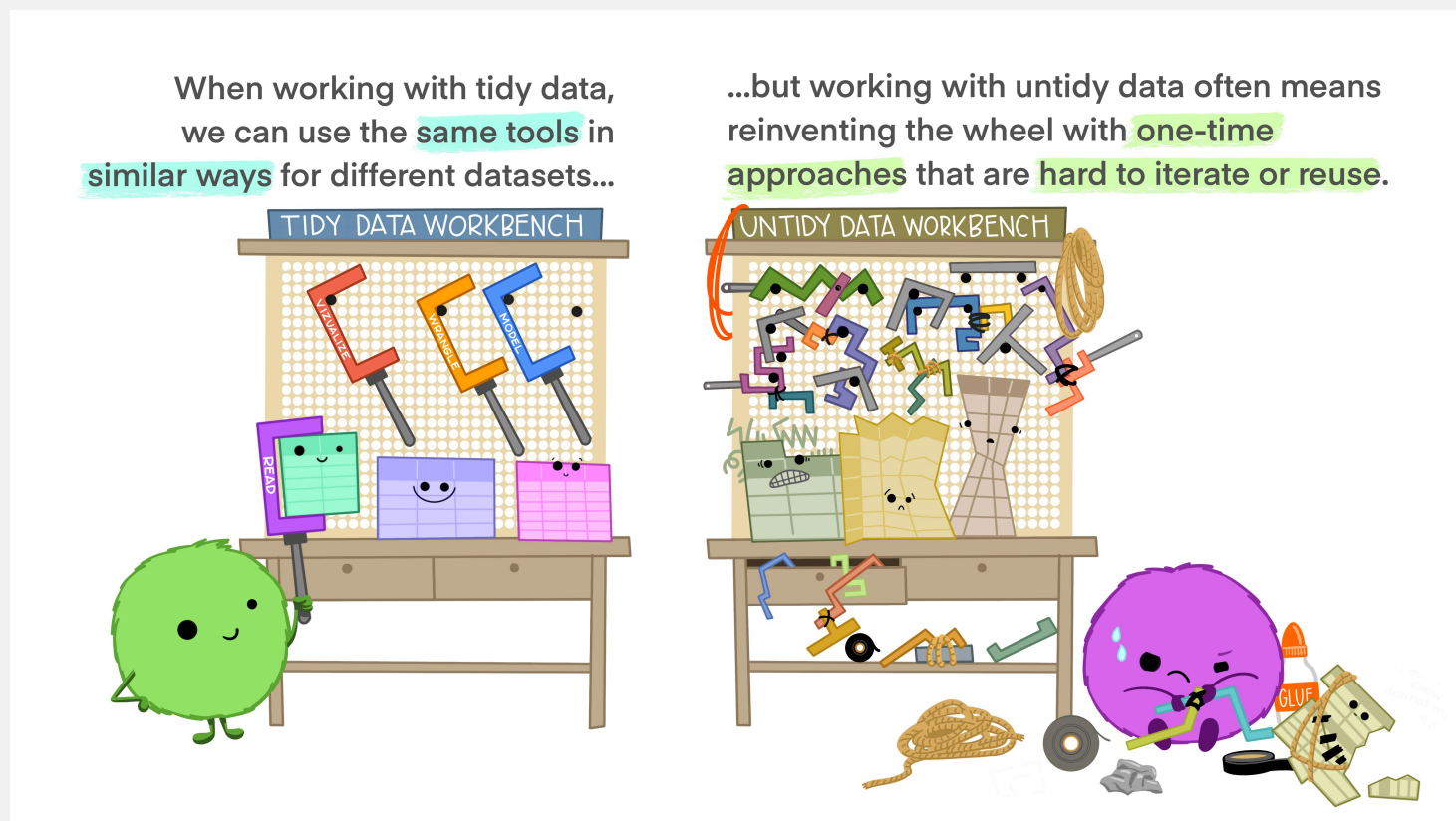


Illustration from the [Openscapes](#) blog *Tidy Data for reproducibility, efficiency, and collaboration* by Julia Lowndes and Allison Horst



Tidy data with tidyr

- **Tidyr** reorganizes the data but **does not change** values
- Most important functionality:
 - **Pivoting**: `pivot_longer` and `pivot_wider`
 - **Splitting** and **combining** columns:
`separate_wider_delim/position/regex` and `unite`
 - Handle **missing values**: `drop_na`, `replace_na`, `complete`

Example



What is not tidy about our weather data?

```
arc_weather
#> # A tibble: 365 × 3
#>   date      foolik_mean_temp_c poolik_mean_temp_c
#>   <date>          <dbl>          <dbl>
#> 1 1988-06-01           8.4           11.8
#> 2 1988-06-02           6            8.90
#> 3 1988-06-03           5.8           8.14
#> 4 1988-06-04           1.8           3.95
#> 5 1988-06-05           6.8           9.81
#> 6 1988-06-06           5.2           8.74
#> 7 1988-06-07           2.2           6.11
#> 8 1988-06-08           9.4          12.1
#> 9 1988-06-09          13.1          16.5
#> 10 1988-06-10          17.7          20.9
#> # i 355 more rows
```

- Each row has multiple observations
- Variables are split across multiple columns

Example



This can be solved with `pivot_longer`

```
pivot_longer(arc_weather,
  cols = c(foolik_mean_temp_c, poolik_mean_temp_c))
#> # A tibble: 730 × 3
#>   date      name      value
#>   <date>    <chr>    <dbl>
#> 1 1988-06-01 foolik_mean_temp_c  8.4
#> 2 1988-06-01 poolik_mean_temp_c 11.8
#> 3 1988-06-02 foolik_mean_temp_c   6
#> 4 1988-06-02 poolik_mean_temp_c  8.90
#> 5 1988-06-03 foolik_mean_temp_c  5.8
#> 6 1988-06-03 poolik_mean_temp_c  8.14
#> 7 1988-06-04 foolik_mean_temp_c  1.8
#> 8 1988-06-04 poolik_mean_temp_c  3.95
#> 9 1988-06-05 foolik_mean_temp_c  6.8
#> 10 1988-06-05 poolik_mean_temp_c  9.81
#> # i 720 more rows
```

Example



You can also directly name the new columns:

```
arc_weather <- pivot_longer(arc_weather,  
  cols = c(foolik_mean_temp_c, poolik_mean_temp_c),  
  names_to = "station",  
  values_to = "mean_temp_c")
```

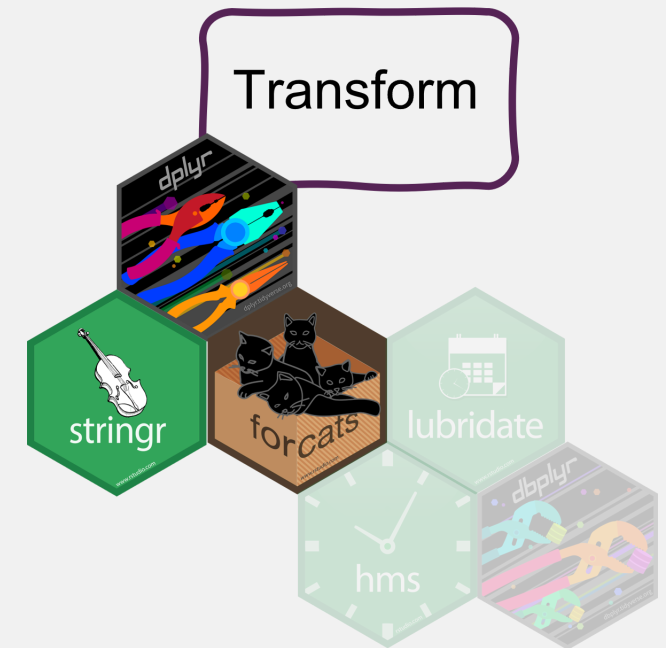
arc_weather

```
#> # A tibble: 730 × 3
```

```
#>   date      station      mean_temp_c  
#>   <date>   <chr>         <dbl>  
#> 1 1988-06-01 foolik_mean_temp_c      8.4  
#> 2 1988-06-01 poolik_mean_temp_c     11.8  
#> 3 1988-06-02 foolik_mean_temp_c      6  
#> 4 1988-06-02 poolik_mean_temp_c     8.90  
#> 5 1988-06-03 foolik_mean_temp_c      5.8  
#> 6 1988-06-03 poolik_mean_temp_c     8.14  
#> 7 1988-06-04 foolik_mean_temp_c      1.8  
#> 8 1988-06-04 poolik_mean_temp_c     3.95  
#> 9 1988-06-05 foolik_mean_temp_c      6.8  
#> 10 1988-06-05 poolik_mean_temp_c     9.81  
#> # i 720 more rows
```

Transform data with dplyr

How to actually change values



Transform data with `dplyr`



- Data cleaning, adding new columns, summarizing, ...
- Depending on the data type in combination with
 - `stringr` for character columns
 - `lubridate` for date-time columns
 - `forcats` for factor columns

Data cleaning and filtering



select picks variables (columns)
based on their names

```
select(arc_weather,  
       date, mean_temp_c)  
#> # A tibble: 730 × 2  
#>   date      mean_temp_c  
#>   <date>      <dbl>  
#> 1 1988-06-01         8.4  
#> 2 1988-06-01        11.8  
#> 3 1988-06-02         6  
#> 4 1988-06-02        8.90  
#> 5 1988-06-03         5.8  
#> 6 1988-06-03        8.14  
#> 7 1988-06-04         1.8  
#> 8 1988-06-04        3.95  
#> 9 1988-06-05         6.8  
#> 10 1988-06-05        9.81  
#> # i 720 more rows
```

filter picks observations (rows)
based on their values

```
filter(arc_weather,  
       station == "foollik_mean_temp_c")  
#> # A tibble: 365 × 3  
#>   date      station      mean_temp_c  
#>   <date>   <chr>      <dbl>  
#> 1 1988-06-01 foollik_mean_temp_c      8.4  
#> 2 1988-06-02 foollik_mean_temp_c       6  
#> 3 1988-06-03 foollik_mean_temp_c      5.8  
#> 4 1988-06-04 foollik_mean_temp_c      1.8  
#> 5 1988-06-05 foollik_mean_temp_c      6.8  
#> 6 1988-06-06 foollik_mean_temp_c      5.2  
#> 7 1988-06-07 foollik_mean_temp_c      2.2  
#> 8 1988-06-08 foollik_mean_temp_c      9.4  
#> 9 1988-06-09 foollik_mean_temp_c     13.1  
#> 10 1988-06-10 foollik_mean_temp_c     17.7  
#> # i 355 more rows
```

Combine **filter** with other packages



lubridate to filter by month

```
# Filter rows from May
filter(arc_weather,
  month(date) == 5)
#> # A tibble: 62 × 3
#>   date      station      mean_temp_c
#>   <date>    <chr>         <dbl>
#> 1 1989-05-01 foolik_mean_temp_c -3.3
#> 2 1989-05-01 poolik_mean_temp_c -0.312
#> 3 1989-05-02 foolik_mean_temp_c -0.8
#> 4 1989-05-02 poolik_mean_temp_c 2.20
#> 5 1989-05-03 foolik_mean_temp_c 0.5
#> 6 1989-05-03 poolik_mean_temp_c 2.59
#> 7 1989-05-04 foolik_mean_temp_c -2.1
#> 8 1989-05-04 poolik_mean_temp_c 0.457
#> 9 1989-05-05 foolik_mean_temp_c -2
#> 10 1989-05-05 poolik_mean_temp_c 1.19
#> # i 52 more rows
```

stringr to filter characters

```
# Filter rows where the station contains "foolik"
filter(arc_weather,
  str_detect(station, "foolik"))
#> # A tibble: 365 × 3
#>   date      station      mean_temp_c
#>   <date>    <chr>         <dbl>
#> 1 1988-06-01 foolik_mean_temp_c 8.4
#> 2 1988-06-02 foolik_mean_temp_c 6
#> 3 1988-06-03 foolik_mean_temp_c 5.8
#> 4 1988-06-04 foolik_mean_temp_c 1.8
#> 5 1988-06-05 foolik_mean_temp_c 6.8
#> 6 1988-06-06 foolik_mean_temp_c 5.2
#> 7 1988-06-07 foolik_mean_temp_c 2.2
#> 8 1988-06-08 foolik_mean_temp_c 9.4
#> 9 1988-06-09 foolik_mean_temp_c 13.1
#> 10 1988-06-10 foolik_mean_temp_c 17.7
#> # i 355 more rows
```

Change and add values



Add columns with `mutate`

```
# Add column with temperature in K
mutate(arc_weather,
  mean_temp_k = mean_temp_c + 274.15)
#> # A tibble: 730 × 4
#>   date      station      mean_temp_c mean_temp_k
#>   <date>    <chr>          <dbl>      <dbl>
#> 1 1988-06-01 foolik_mean_temp_c      8.4      283.
#> 2 1988-06-01 poolik_mean_temp_c     11.8      286.
#> 3 1988-06-02 foolik_mean_temp_c      6       280.
#> 4 1988-06-02 poolik_mean_temp_c     8.90      283.
#> 5 1988-06-03 foolik_mean_temp_c      5.8      280.
#> 6 1988-06-03 poolik_mean_temp_c     8.14      282.
#> 7 1988-06-04 foolik_mean_temp_c      1.8      276.
#> 8 1988-06-04 poolik_mean_temp_c     3.95      278.
#> 9 1988-06-05 foolik_mean_temp_c      6.8      281.
#> 10 1988-06-05 poolik_mean_temp_c     9.81      284.
#> # i 720 more rows
```

Change and add values



Change the values of existing columns with `mutate`

```
# Remove _mean_temp_c from station
arc_weather <- mutate(arc_weather,
                      station = str_remove(station, "_mean_temp_c"))

arc_weather
#> # A tibble: 730 × 3
#>   date      station mean_temp_c
#>   <date>    <chr>         <dbl>
#> 1 1988-06-01 foolik          8.4
#> 2 1988-06-01 poolik         11.8
#> 3 1988-06-02 foolik          6
#> 4 1988-06-02 poolik         8.90
#> 5 1988-06-03 foolik          5.8
#> 6 1988-06-03 poolik         8.14
#> 7 1988-06-04 foolik          1.8
#> 8 1988-06-04 poolik         3.95
#> 9 1988-06-05 foolik          6.8
#> 10 1988-06-05 poolik         9.81
#> # i 720 more rows
```

Summarize data



`summarize` will collapse the data to a single row

```
# Calculate the overall mean temperature
summarize(arc_weather,
           overall_mean = mean(mean_temp_c, na.rm = TRUE))
#> # A tibble: 1 × 1
#>   overall_mean
#>   <dbl>
#> 1      -7.10
```

Summarizing data by group



`summarize` in combination with `group_by` will collapse the data to a single row per group

```
# Calculate the mean temperature for each station
arc_weather_group <- group_by(arc_weather, station)
summarize(arc_weather_group,
  overall_mean = mean(mean_temp_c, na.rm = TRUE))
#> # A tibble: 2 × 2
#>   station overall_mean
#>   <chr>         <dbl>
#> 1 foolik      -8.60
#> 2 poolik      -5.60
```

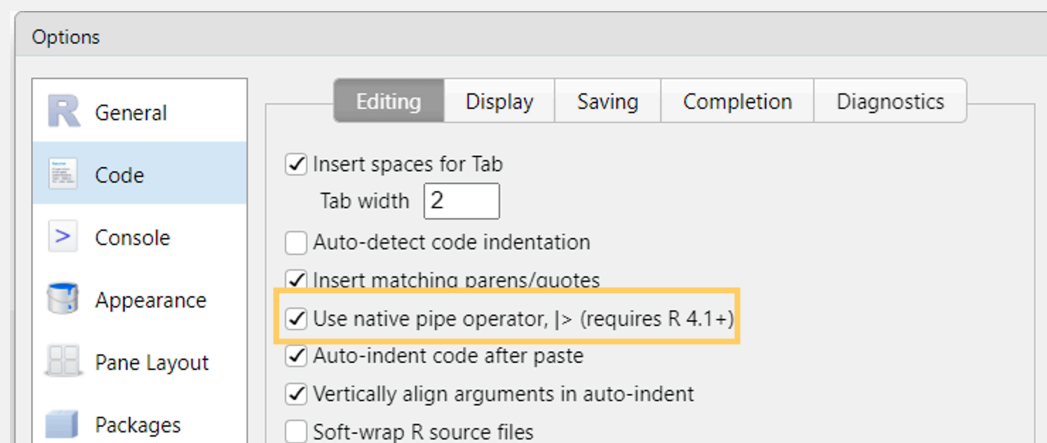
- `group_by` can be used for any dplyr function to perform the operation by group

The pipe |>

Data transformation often requires **multiple operations** in sequence.

The pipe operator |> helps to keep these operations clear and readable.

- You may also see %>% from the **magrittr** package (tidyverse)
- Turn on the native R pipe |> in **Tools -> Global Options -> Code**



The pipe |>

Let's look at an example without the pipe:

```
arc_weather_group <- group_by(arc_weather, station)
summarize(arc_weather_group,
          overall_mean = mean(mean_temp_c, na.rm = TRUE))
```

The pipe operator makes it very easy to combine these operations

```
arc_weather |>
  group_by(station) |>
  summarize(overall_mean = mean(mean_temp_c, na.rm = TRUE))
```

You can read from top to bottom and interpret the |> as an *"and then do"*.

The pipe |>

But what is happening?

The pipe is “pushing” the result of one line into the first argument of the function from the next line.

```
arc_weather |>
  group_by(station)

# instead of
group_by(arc_weather, station)
```

Piping works perfectly with the **tidyverse** functions because they are designed to **return a tibble** and **take a tibble as first argument**.

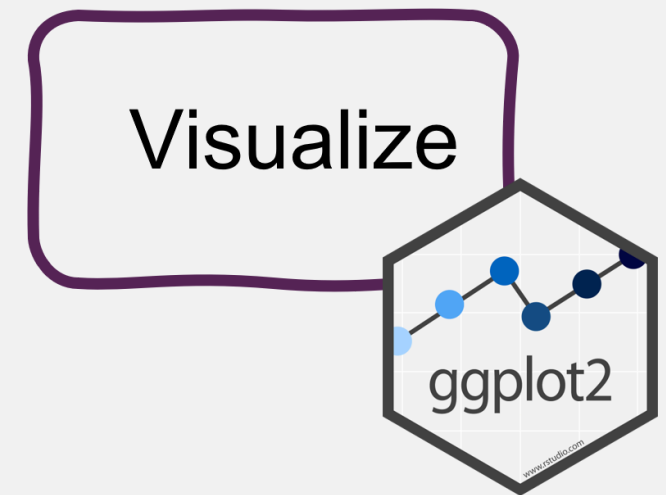


Tip

Use the keyboard shortcut **Ctrl/Cmd + Shift + M** to insert |>

Visualize data with ggplot2

Quick exploratory plots and data masterpieces

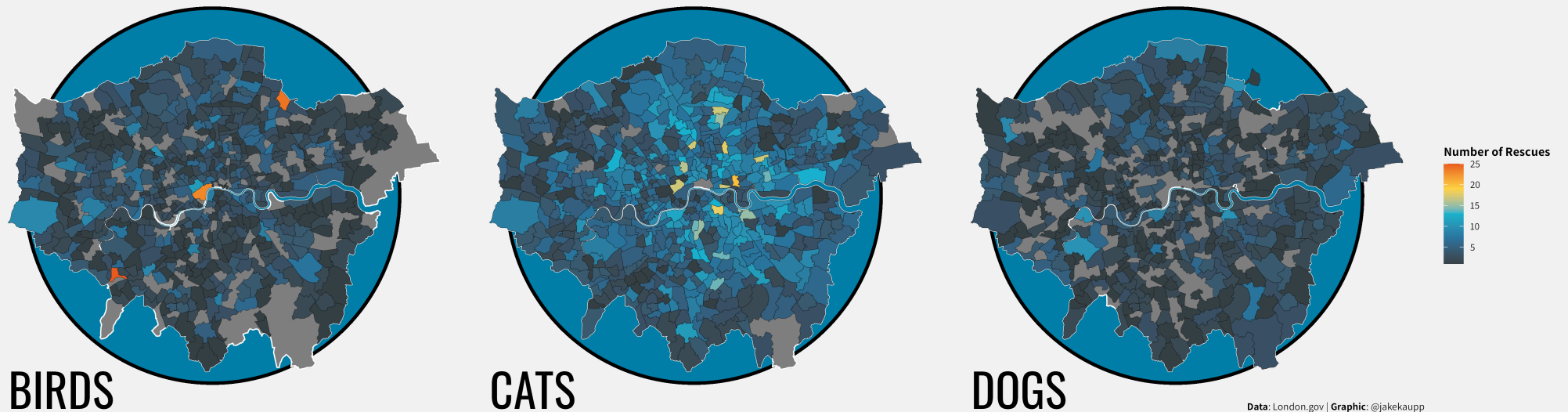


A ggplot showcase



Frequency of Rescues of Birds, Cats and Dogs in London from 2009-2021

Illustrated below in three choropleth maps are rescues of birds, cats and dogs in London wards. Darker colors indicate lower rescue numbers while brighter colors indicate a greater number of rescues in that ward.

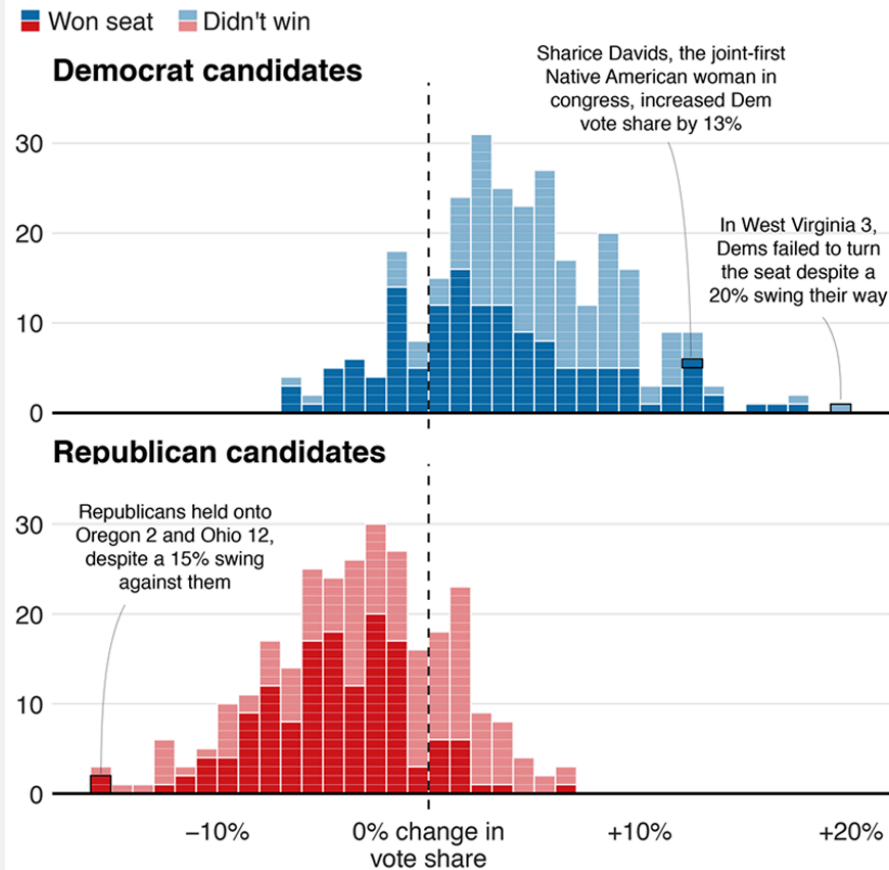


Visualization by [Jake Kaupp](#), code available on [Github](#)

A ggplot showcase



Blue wave

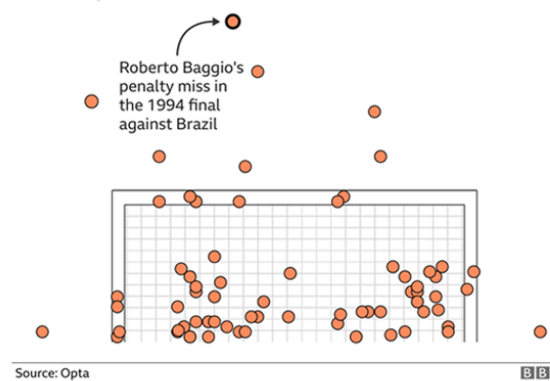


Source: AP, 19:01 ET

BBC

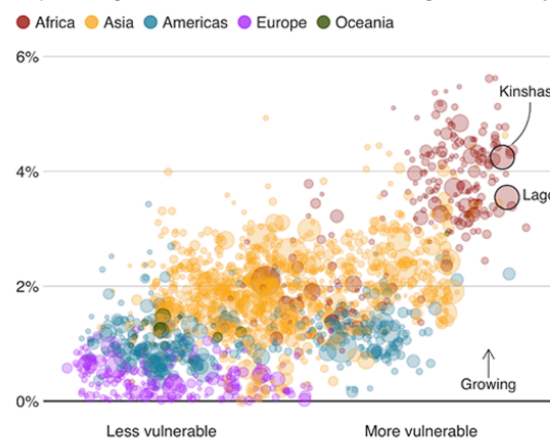
Where penalties are saved

World Cup shootout misses and saves, 1982-2014



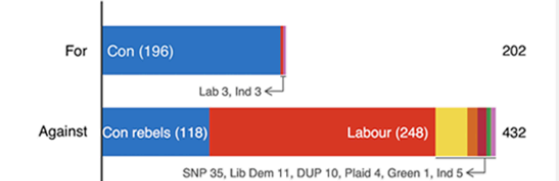
Fast-growing cities face worse climate risks

Population growth 2018-2035 over climate change vulnerability



Source: Verisk Maplecroft. Circle size represents current population.

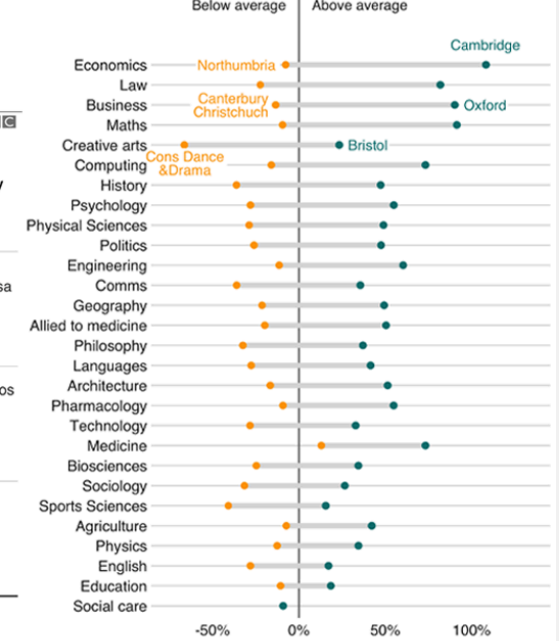
MPs rejected Theresa May's deal by 230 votes



Source: Commons Votes Services. Excludes 'tellers', the Speaker and deputies

Earnings vary across units even within subjects

Impact on men's earnings relative to the average degree



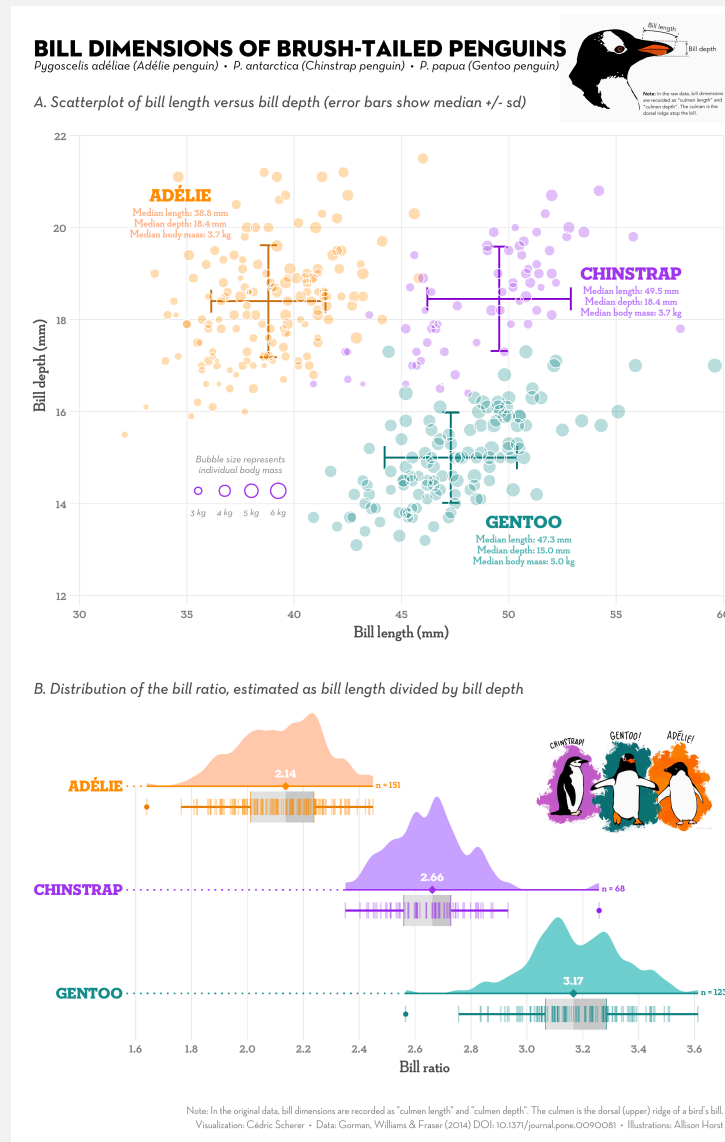
Source: Institute for Fiscal Studies

BBC

Visualizations produced by the BBC News data team

Selina Baldauf // Tidyverse

A ggplot showcase



Visualization by Cédric Scherer, code available on [GitHub](#) and the [tidyverse](#)

Advantages of ggplot



- **Consistent** grammar
- **Flexible** structure allows you to produce any type of plots
- Highly **customizable appearance** (themes)
- Many **extension packages** that provide additional plot types, themes, colors, animation, ...
 - Find a list of ggplot extensions [here](#)
- Perfect package for **exploratory data analysis** and **beautiful plots**

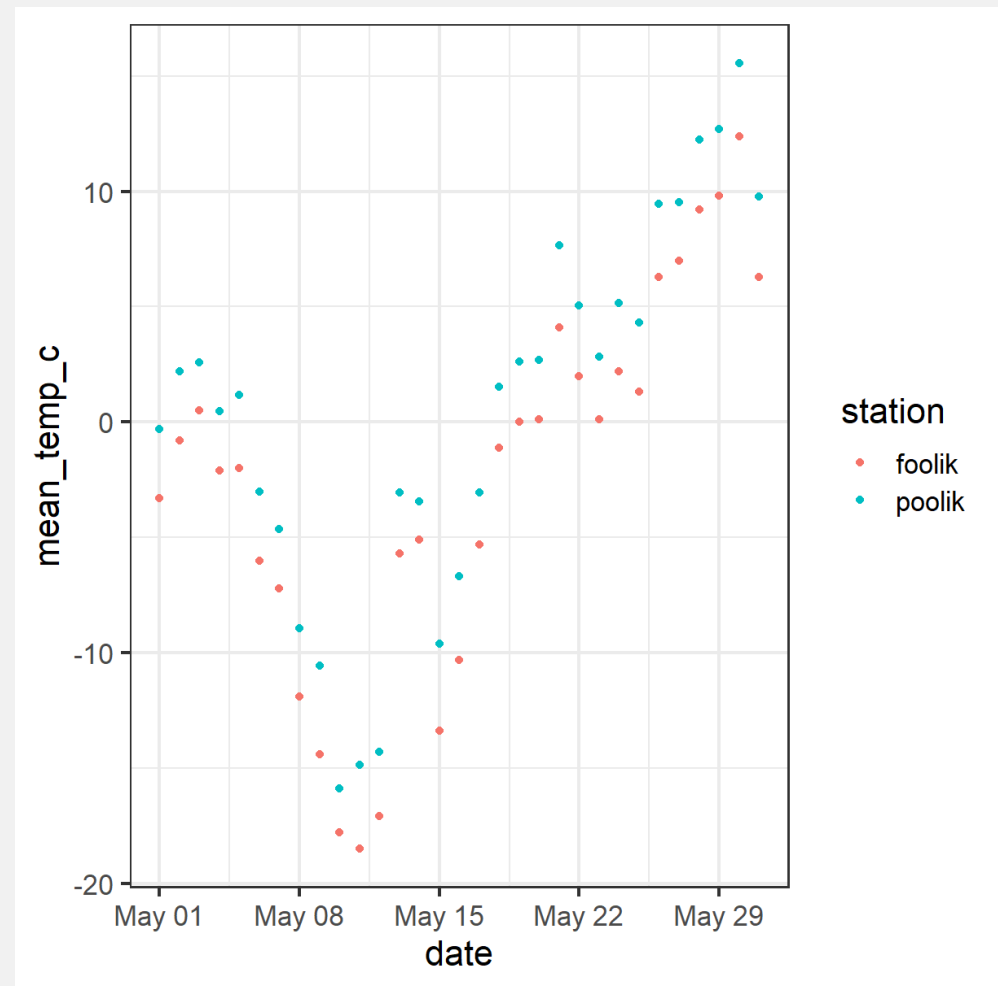
Basic ggplot



Basic idea: Stack layers of the plot on top of each other with +

```
arc_weather_may <- filter(arc_weather,  
  month(date) == 5)  
ggplot(data = arc_weather_may,  
  aes(  
    x = date,  
    y = mean_temp_c,  
    color = station  
  )) +  
  geom_point()
```

- **aesthetics** define how data variables are mapped onto plot properties
- **geoms** define the type of plot (e.g. points, lines, bars, ...)

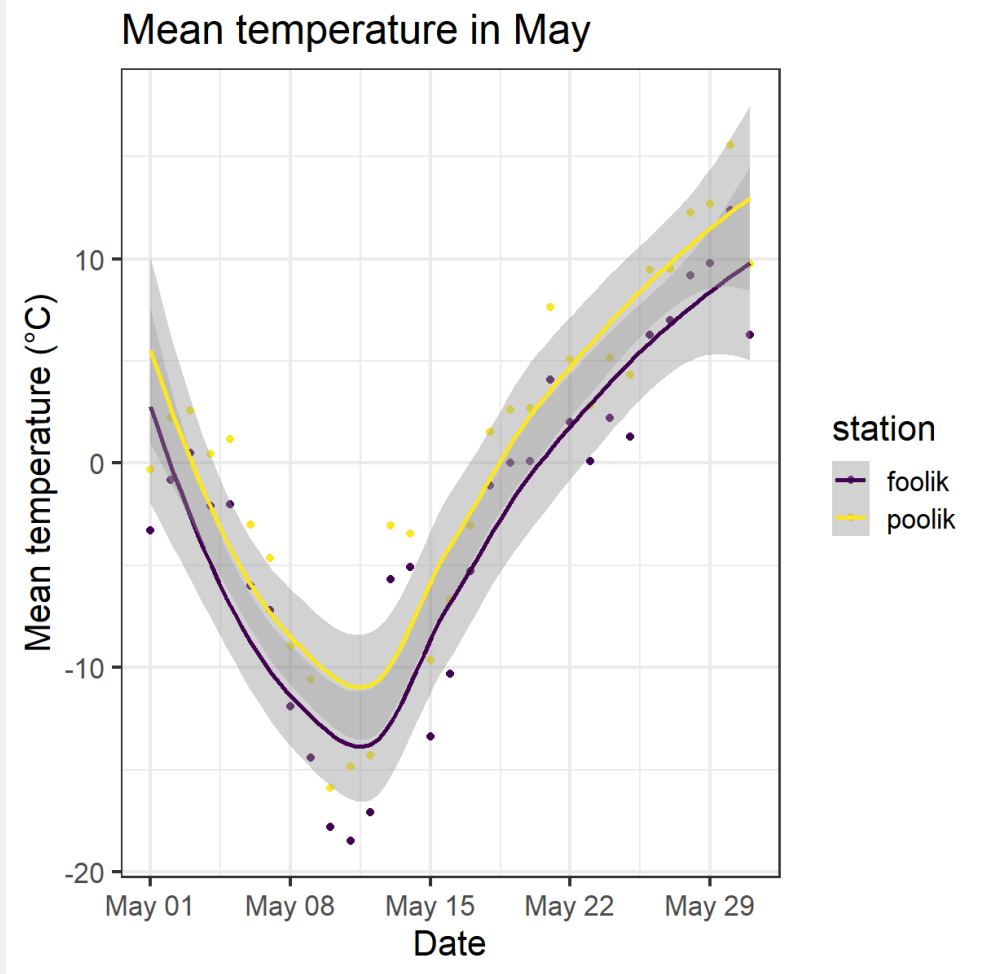


Basic ggplot



- Gplot offers many more customization options and layers
- Works well with the pipe |>

```
arc_weather |>
  filter(month(date) == 5) |>
  ggplot(aes(
    x = date,
    y = mean_temp_c,
    color = station)) +
  geom_point() +
  geom_smooth() +
  labs(
    title = "Mean temperature in May",
    x = "Date",
    y = "Mean temperature (°C)") +
  scale_color_viridis_d()
```



Functional programming with purrr



Functional programming

Functional programming with purrr



- Apply functions to multiple elements of a list/vector/...
 - Replace e.g. for loops/apply-functions
 - Often **purrr** functions are more intuitive
 - See [here](#) for a full comparison between base R functions and purr functions
- Most important function: **map**
 - Comes in different versions, depending on the input and the desired output

A simple example



Draw 4 numbers from normal distribution with means 1, 2, 3, 4.

Do it by hand

```
# Do it by hand
set.seed(123)
rnorm(n = 4, mean = 1, sd = 1)
#> [1] 0.4395244 0.7698225 2.5587083 1.0705084
rnorm(n = 4, mean = 2, sd = 1)
#> [1] 2.1292877 3.7150650 2.4609162 0.7349388
rnorm(n = 4, mean = 3, sd = 1)
#> [1] 2.313147 2.554338 4.224082 3.359814
rnorm(n = 4, mean = 4, sd = 1)
#> [1] 4.400771 4.110683 3.444159 5.786913
```

Use purrr

```
# Use purrr
set.seed(123)
means <- 1:4
samples <- map(means, rnorm, n = 4, sd = 1)
str(samples)
#> List of 4
#> $ : num [1:4] 0.44 0.77 2.56 1.07
#> $ : num [1:4] 2.129 3.715 2.461 0.735
#> $ : num [1:4] 2.31 2.55 4.22 3.36
#> $ : num [1:4] 4.4 4.11 3.44 5.79
```

Different versions of the **map** function



These are just some examples, there are **many more options**:

Output	Input	Purrr function
List	1 Vector	map
List	2 Vectors	map2
Vector of desired type	1 Vector	map_lgl
Vector of desired type	2 Vectors	map2_lgl
Data frame	1 Vector	map_df
Data frame	2 Vectors	map2_df

A more complex example



Read in and analyse multiple files with the same structure at once.

```
# Step 1: List all files in your data directory
paths <- list.files("data")
```

```
#> [1] "foolik.csv" "poolik.csv"
```

```
# Step 2: Read all files into a list of tibbles
arc_weather_files <- map(paths, read_csv)
arc_weather_files
```

```
#> [[1]]
```

```
#> # A tibble: 365 × 3
```

```
#>   date      station mean_temp_c
```

```
#>   <date>    <chr>      <dbl>
```

```
#> 1 1988-06-01 foolik      8.4
```

```
#> 2 1988-06-02 foolik      6
```

```
#> 3 1988-06-03 foolik     5.8
```

```
#> 4 1988-06-04 foolik     1.8
```

```
#> 5 1988-06-05 foolik     6.8
```

```
#> 6 1988-06-06 foolik     5.2
```

```
#> 7 1988-06-07 foolik     2.2
```

```
#> 8 1988-06-08 foolik     9.4
```

```
#> 9 1988-06-09 foolik    13.1
```

```
#> 10 1988-06-10 foolik    17.7
```

```
#> # i 355 more rows
```

```
#>
```

```
#> [[2]]
```

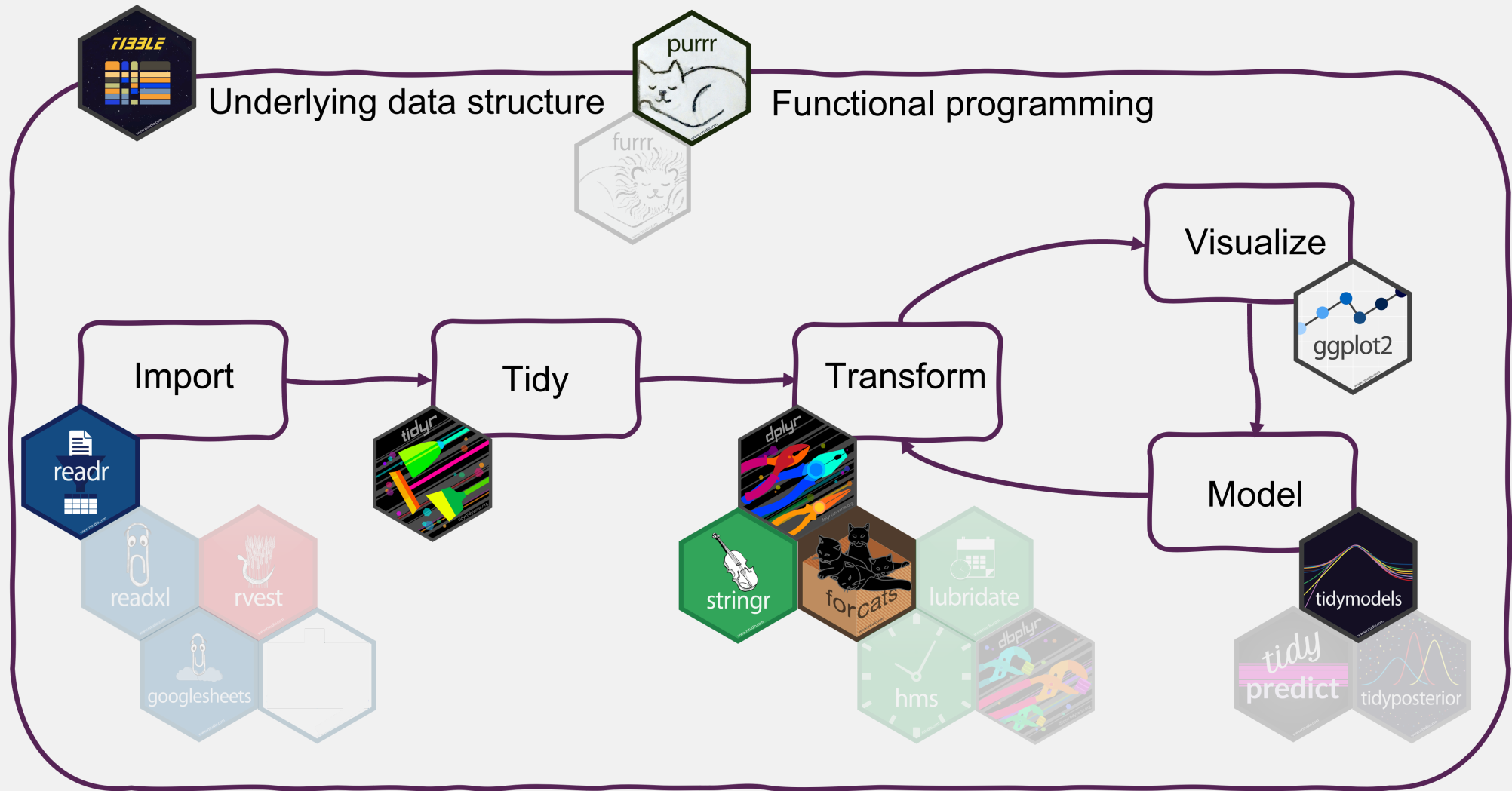
A more complex example



Read in and analyse multiple files with the same structure at once.

```
# Step 3: Apply a linear model to each station file,  
# get the model summary of each model and extract the coefficients  
arc_weather_files |> # Take the data  
  map(\(x) lm(mean_temp_c ~ date, data = x)) |> # Apply the same model to every tibbel in the list  
  map(summary) |> # Get the model summary of both models  
  map_dbl("r.squared") # Extract the r-squared from the model summaries  
#> [1] 0.2018279 0.2000210
```

Summary



Summary

Tidyverse makes data analysis

- efficient
- intuitive
- consistent

Tidyverse code is more **readable** and **easier to write & maintain**.

This facilitates



Learning



Collaboration



Reproducible research

Summary

Where to get started with the tidyverse?

- Check out the [lecture website](#) for links to documentation, cheatsheets and book

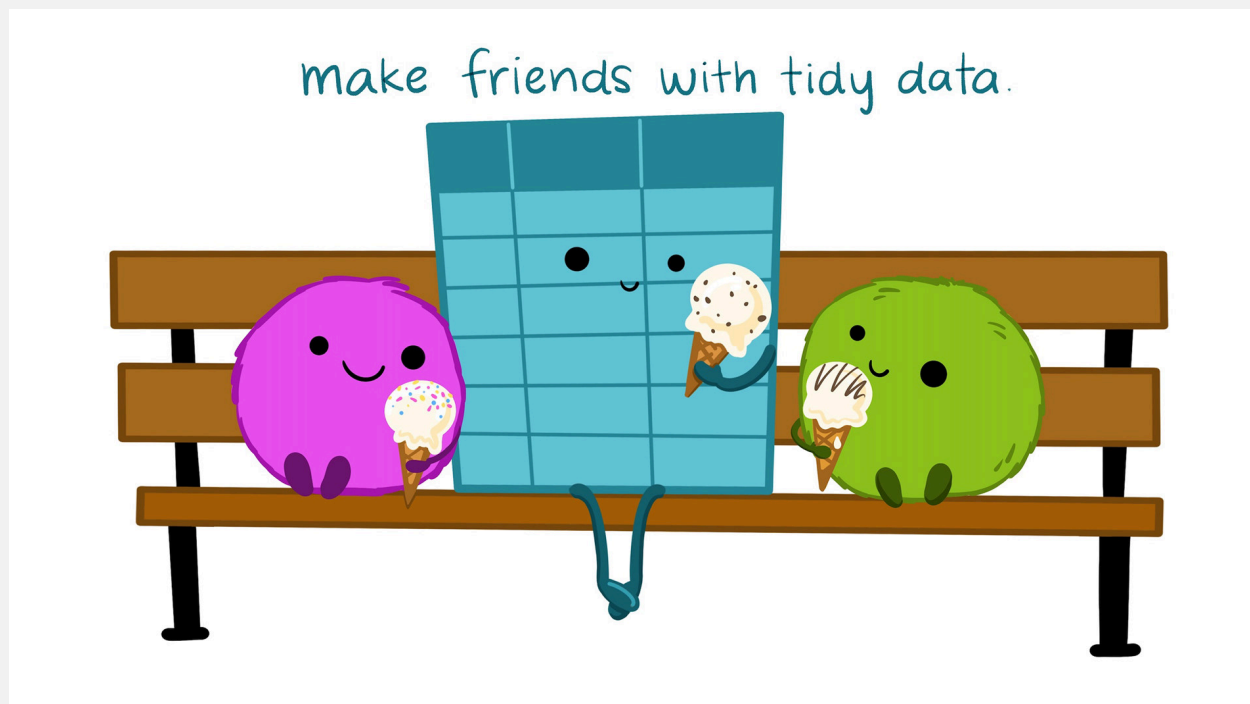


Illustration from the [Openscapes](#) blog *Tidy Data for reproducibility, efficiency, and collaboration* by Julia Lowndes and Allison Horst

Selina Baldauf // Tidyverse

Next lecture

Semester break in March!

Topic t.b.a.

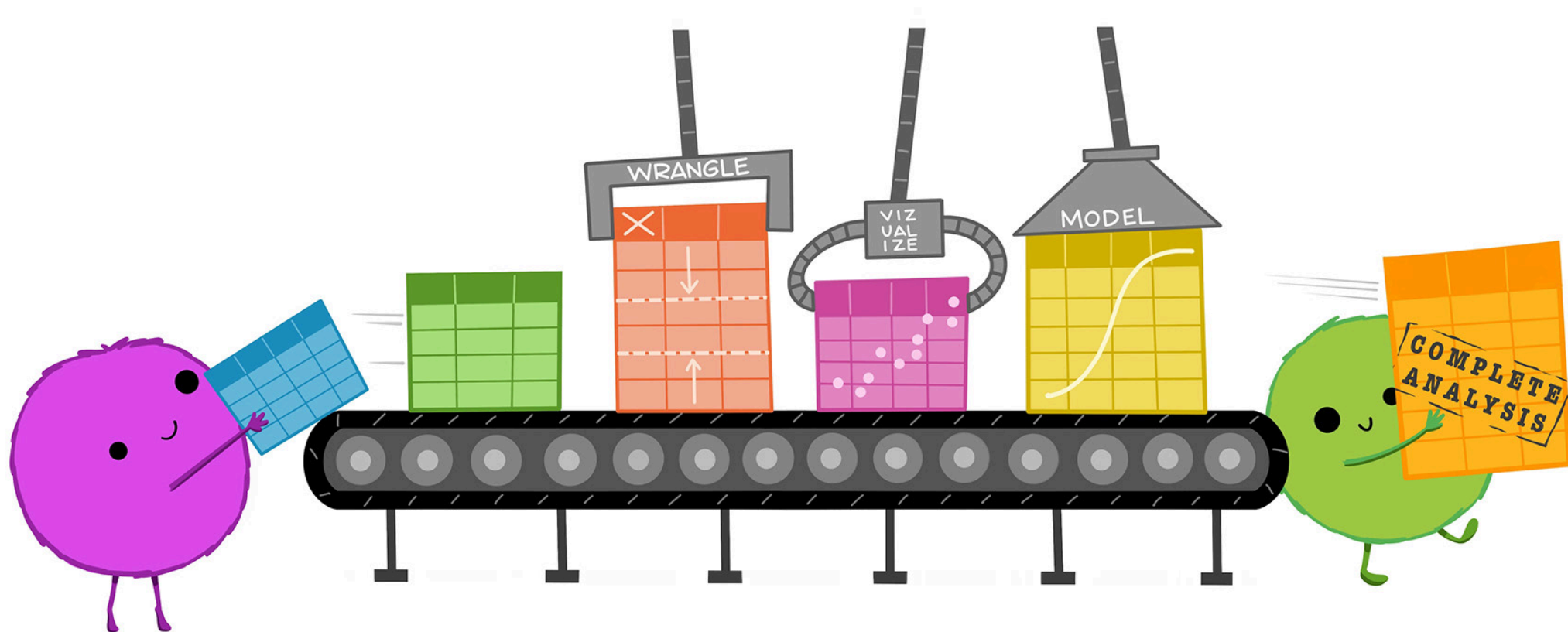
 18th April  4-5 p.m.  Webex

 Subscribe to the mailing list

 For topic suggestions and/or feedback [send me an email](#)

Thank you for your attention :)

Questions?



References

- [Tidyverse website](#): find links to all package documentations
- [Cheatsheets](#)
- [R for Data Science](#) book: Learn data analysis with the tidyverse